Project Part 1

Thomas Kennedy & Cathy Le

Team GitHub Repository: https://github.com/tksoftw/stat-3080-project-1

Introduction

Capital Bikeshare stations are a familiar sight across Washington, D.C. and the surrounding metropolitan counties. As these bright red bikes become an increasingly popular way to navigate the city, they naturally alter the flow of daily activity in densely populated neighborhoods. This shift raises a practical question for urban planners, local businesses, and everyday residents regarding neighborhood safety. People want to know if placing a mini-transit hub on their block changes the local crime rate, and whether these hubs make the immediate surroundings safer or more vulnerable.

To get a clear answer, our research focuses on data from Washington, D.C. in 2026. We want to know if proximity to a Capital Bikeshare station correlates with higher rates of property crime but lower rates of violent crime. We examine these two categories separately because human behavior drives them in very different directions. A busy docking station creates a centralized area of foot traffic. For property crimes like theft or vandalism, that constant churn of people might provide a convenient, target-rich environment. Conversely, a steady stream of riders creates a layer of natural surveillance. Having more eyes on the street and constant pedestrian movement might actively discourage violent crimes from occurring in the immediate area.

Data Summary

We use two open datasets from the D.C. government, scoped to Washington, D.C. in 2026. Because they include all reported incidents and all active stations in 2026, they represent population data for reported crimes.

- **Capital Bikeshare Locations**: Contains 819 active stations, with `LATITUDE` and

`LONGITUDE` coordinates.

- **Crime Incidents in 2026**: Contains 4,801 reported offenses with location data and `OFFENSE` category.

A crime incident is labeled "near" a bikeshare station if it falls within approximately 310 meters, or a 5-block radius, of any station. Otherwise, it is "far". Incidents are also classified as "Property" such as theft and burglary, or "Violent" such as robbery and homicide.

Data Limitations: The dataset contains only *reported* crimes; minor property thefts are likely underreported, especially in certain neighborhoods. Furthermore, the GPS coordinates of incidents represent approximate block-level locations rather than exact addresses, introducing some measurement error in the distance calculations. Additionally, some Capital Bikeshare locations were located far outside D.C. in the surrounding suburbs, so we filtered out any stations outside a 12 km radius from the geographic center of D.C. Despite these limitations, the data are appropriate to answer our research question because they represent the most comprehensive geographic record of urban crime and transit behavior available for our scope.

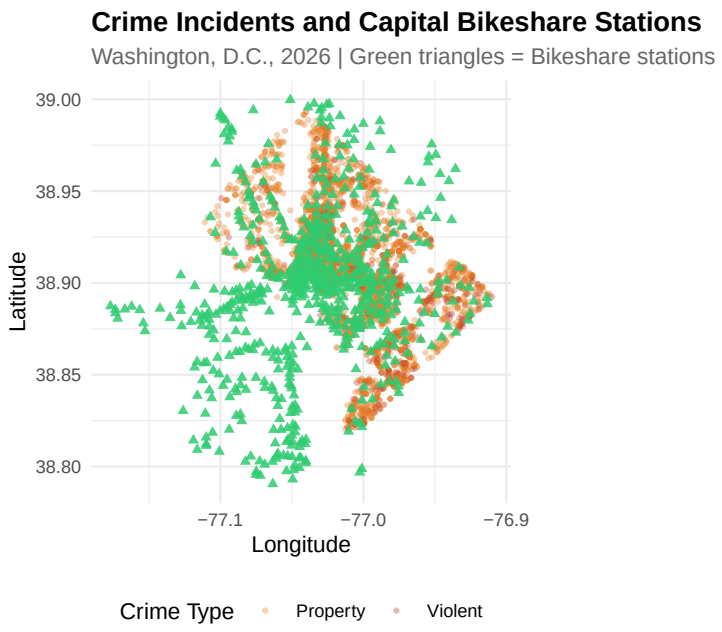
Exploratory Analysis

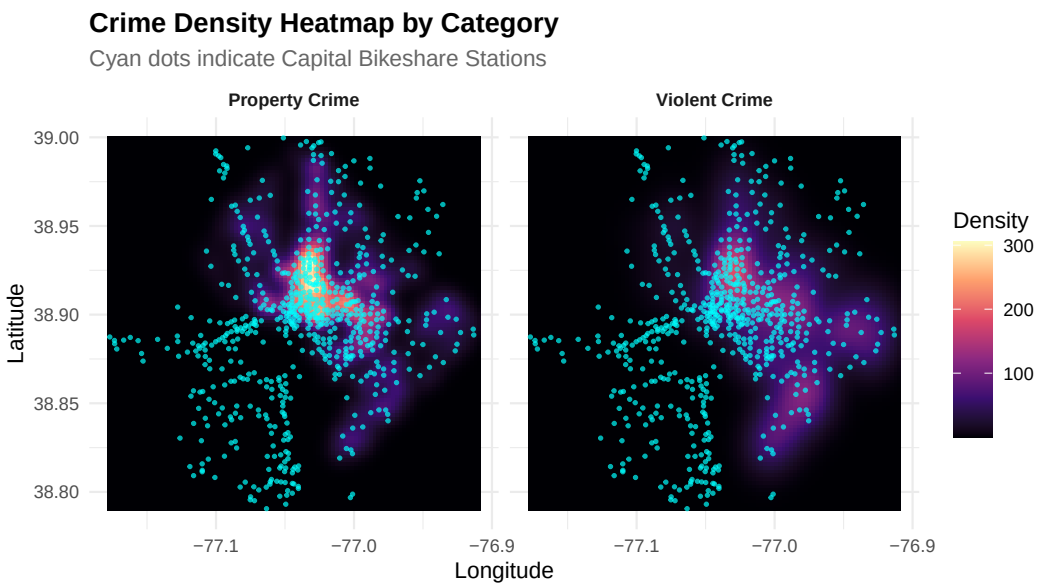
```
## Numerical Summary 1: Incident Counts by Proximity and Crime Category

##          Category
## Proximity Property Violent
##      Far       908     185
##      Near     3295     413

## Numerical Summary 2: Crime Category Proportions Within Proximity Group (%)

##          Category
## Proximity Property Violent
##      Far      83.1    16.9
##      Near     88.9    11.1
```





Conclusions

The 4,801 incidents in our dataset show a shift in offense types based on how close you are to a docking location. Near the stations, property crimes dominate the environment, making up 88.9% of reported incidents compared to 11.1% for violent crimes. Move away from the bikeshare network, and that ratio changes. In those outer zones, the proportion of property crimes falls to 83.1%, while violent crimes rise to 16.9%.

The two maps we generated earlier show exactly how this geography plays out on the streets. On the dot map, green triangles mark the bikeshare stations packed into D.C.'s central core, and the orange dots tracking property crimes stack heavily over those exact same blocks. The density heatmap brings the contrast into sharper focus. Both categories of crime share a center in the middle of the city, but the property crime hotspot remains intensely concentrated along the bikeshare network's footprint. Violent crime scatters much more broadly, spreading into outer neighborhoods that lack docking infrastructure entirely.

Nevertheless, a major underlying factor drives this visual overlap: population density. Urban planners intentionally build bikeshare infrastructure in D.C.'s most densely populated and commercially active neighborhoods. These high-density blocks naturally generate more opportunities for theft simply because they contain more retail spaces, more pedestrians, and more unattended belongings. The concentrated property crime we see around the bikeshare network might be a byproduct of the sheer volume of daily human activity in the city's core, rather than the docking stations themselves acting as the primary draw for offenses.

References

Crime Incidents in 2026. (n.d.). [Dataset]. Link to source

Capital Bikeshare locations. (n.d.). Link to source

Rimolo, S. (2023, April 20). Bicycle share systems: a predictor of crime? Link to source

Google. (2026). Gemini (March 8 version) [Large language model]. Link to source

Anthropic. (2026). Claude (Version 4.6 Opus) [Large language model]. Link to source
Anthropic. (2026). Claude (Version 4.6 Sonnet) [Large language model]. Link to source
Google. (2026). Gemini (Version 3.1 Pro) [Large language model]. Link to source